

Systems for Design and Implementation : Assignment 4

WE CANNOT RUN THE LABS WITHOUT:

-the server running locally and being accessible on the LAN

-if you deployed the server on vercel for the previous lab,

change the server to run on the LAN

SILVER CHALLENGE

- implement proper authentication and authorization to ALL THE ROLES established in the previous assignment
 - generate tokens for different permission schemes
- manage sessions
- implement 3 way authentication and password recovery capabilities

GOLD CHALLENGE

HACK THE PLANET!

In the lab you will hack the other's system with whichever hacking tool or trick you have up your sleeve. The mission, should you choose to accept it, is the following:

learn at least two tools:

- wireshark [Wireshark • Go Deep | Download](#) for packet sniffing and understanding the wireless information flow on the network
- jmeter [Apache JMeter - Apache JMeter™](#) it is traditionally used for testing the backend but we will use it to DDOS your system

but of course it is preferable you learn any other cool hacking tool and share it with the class in order to sniff any vulnerability your colleague's implementation has and exploit it. Also, you need to protect your own fort by implementing defense mechanisms for any type of attack you might do upon others. You are both the PREDATOR and the PREY.

Deadline for the bronze[5p] challenge: Week 10

Deadline for the silver[+2p] and gold[+3p] challenges without delay: Week 12[max grade 10]

Deadline for the silver[+1p] and gold[+1.5p] challenges with one delay: Week 13[max grade 7.5]

Deadline for the silver[+0.5p] and gold[+0.75p] challenges with two delays: Week 14[max grade 6.25]

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Use local LLMs (free versions only, don't spend your money) and use them locally to find vulnerabilities in the products of your colleagues.

Minimum implementation on the defensive side to get the gold:

- implement a functionality that provides a statistic based on a many-to-many relationship that is heavily computational, fill all the tables with a huge amount of proper information generated with faker [Getting Started | Faker](#) and then optimize it as much as you can using indices on the database or data cacheing methods, whichever works to decrease data fetching latency. show the improvement in performance using jmeter. illustrate how the naive implementation succumbs to DDOS attempts whereas the improvement is more robust
- use AI to monitor suspicious user behaviour based on the infrastructure built for the previous assignment. simulate suspicious behaviour of clients using [Fast and reliable end-to-end testing for modern web apps | Playwright](#) or other UI robots and show how your system detects their behaviour using AI models in real-time

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